Day 63 - Simulation Environment and Test Candidate Profiles

End-to-end simulation environment and profile summary. The demo dataset is supplied separately.

Dataset organization: **data/day63_demo/dataset_manifest.json** indexes two separate job descriptions, four individual candidate profile files, and one shared machine-test task.

Simulation environment

The Day 63 runner processes fictional candidate records through the existing Zecpath AI pipeline. It demonstrates two Python job descriptions - mid-level backend and associate API - with four profiles assigned to the suitable opening. The report records each stage score, final recommendation, and observed scoring limitations.

Pipeline stages

1. **Resume intake and consent:** load a structured synthetic resume record and grant the demo consent types. The file-upload endpoint and resume parser are not exercised.
2. **ATS:** score skills, experience, education, and semantic similarity against the assigned job description.
3. **Screening:** progress through screening questions and score supplied answers with the configured screening rubric.
4. **HR interview:** run the interview engine with profile-specific fictional responses and its deterministic communication and behavioral scoring.
5. **Technical interview:** capture answers through the technical question flow; use the stated rubric fixture for the numeric score.
6. **Machine test and decision:** score the supplied test rubric, combine all five round scores, apply eligibility rules, and generate a final recommendation.

Run from the repository folder

```
.\venv\Scripts\Activate.ps1
python main_day63_end_to_end_simulation.py
```

The runner writes the results to reports/day63_end_to_end_simulation.json. Model initialization may take longer the first time, depending on the local model cache.

Test candidate profiles and observed results

Alex Morgan - Python Backend Developer - Mid-Level (3.5 years)
Strong, detailed resume with directly relevant backend experience.
ATS 95.59; screening 95.00; HR 90.12; technical 87.20; machine test 92.40; unified 91.83.
Recommendation: **Selected**.

Riley Chen - Python API Developer - Associate (1.5 years)
Adequate early-career profile with relevant project evidence and a few missing preferred tools.
ATS 94.71; screening 85.57; HR 89.36; technical 71.25; machine test 70.80; unified 82.82.
Recommendation: **Selected**.

Taylor Brooks - Python Backend Developer - Mid-Level (2 years)
Career changer with a nontraditional education path and relevant self-directed/volunteer work.
ATS 67.91; screening 87.43; HR 87.93; technical 78.40; machine test 78.80; unified 79.10.
Recommendation: **Selected**.

Jamie Rivera - Python API Developer - Associate (0.8 years)
Sparse entry-level profile with basic exposure, limited evidence, and an explicit skill/experience gap.
ATS 70.60; screening 82.04; HR 89.40; technical 42.40; machine test 31.00; unified 63.09.
Recommendation: **Hold / Review**.

Observed scoring behavior

HR interview scores are tightly clustered across varied profiles. Observed range: 87.93 to 90.12 (spread 2.19). The deterministic HR heuristics produce similar outputs even when resume and technical evidence differ. Treat this stage as a conversation aid, not a reliable ranking signal.

A limited-evidence profile receives a relatively high screening score. DAY63-SYNTH-004: 82.04. The configured screening rubric also rewards clarity and consistency, so its aggregate score can remain high when job-skill evidence is limited. Compare it with ATS, interview, and human-reviewed evidence.

Scope and review

All profiles and responses are fictional. Screening intent classification is bypassed in this fixture run. Technical scores and machine-test pass counts come from supplied rubric fixtures; candidate code is captured but not executed. The recommendations are demonstration outputs and require human review before any real hiring decision.